

Ayan-Yue Gupta

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Computational sociologist working at the intersection of NLP, computational social science, and political sociology. Research focuses on applying computational text analysis with large language models and network methods to political discourse (including policy discourse, misinformation/conspiracies, news media discourse). Particular theoretical interests include pragmatist philosophy of language, discourse analysis, and more generally the relation between language and politics.

Education

PhD Sociology – Thesis Title: *BERT for Discourse Analysis: A Pragmatist Approach to Governmentality* September 2018 – March 2023

University of Bristol – School of Sociology, Politics and International Studies

- Supervisors: Prof. Thomas Osborne, Dr. Leonidas Tsilipakos, Prof. Gregor McLennan

MA Fine Art

University College London – Slade School of Fine Art

September 2014 – September 2016

BA Art History and Philosophy

University of Leeds

September 2011 – July 2014

Work Experience

Postdoctoral Fellow (Contrattista di Ricerca)

University of Milan – Behave Lab, Department of Social and Political Sciences

December 2025 – Present

Working on the [IANUS](#) project, using large language models and network analysis to examine the impact of MSCA research on EU Missions policy.

Adjunct Lecturer

University of Milan – Department of Social and Political Sciences

4 May 2026 – 9 June 2026

Designed and led the module *Text as Data: Computational Text Analysis for the Social Sciences*. Taught MA, PhD and postdocs applications of NLP methods to social science in Python and R, covering text embeddings, topic modelling, classification, large language models and data collection.

Postdoctoral Fellow (Assegno di Ricerca)

University of Milan – Behave Lab, Department of Social and Political Sciences

May 2024 – November 2025

Worked on the [ConsNet](#) project, using large language models to analyse the evolution of health-related conspiracy theories on

social media.

Workshop Leader

May 2025

University of Milan – Behave Lab, Department of Social and Political Sciences

Lead sessions teaching basics of computational text analysis with Python, as part of the Behave Lab's [Doing Computational Social Science with Python and R](#) workshop series.

Graduate Teacher

September 2019 – June 2024

University of Bristol – School of Sociology, Politics and International Studies

Modules taught:

- Introduction to the Sociology of Culture
 - Thinking Sociologically (Introductory sociology module)
 - Doing Social Research (Ethics and methodology module)
 - Political Concepts (Political philosophy module)
 - Thinking Politically (Introductory politics module)
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Graduate Teacher

January 2023 – September

University of Bath – Department of Politics, Languages and International Studies

Modules taught:

- International Political Economy
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Books

Gupta, A. (2025). *The Pragmatics of Governmental Discourse: Resilience, Sustainability and Wellbeing*. Routledge: London. [ISBN: 9781032777511](#)

Gupta, S., Allen, R., Basu, M., Durão, F., **Gupta, A.**, Katsarka, M., Schuller, S., Seed, J., Tu, P. (2020). *Social Analysis and the Covid-19 Crisis*. Routledge: London. [ISBN: 9780367636616](#)

Papers

Gupta, A., Dimitri, G., Vacca, R. (2026). Semantic Structure in Large Language Model Disagreement: An Analysis of Representation Spaces in Conspiracy Content Classification. *Proceedings of the 12th IAPR TC3 Workshop on Artificial Neural Networks in Pattern Recognition*.

Gupta, S. & **Gupta, A.** (2025). Catchwords: An Essay on Research Methods. *Keywords: A Journal of Cultural Materialism*, 22, pp. 70-96. Available at: <https://oro.open.ac.uk/100561/>

Gupta, A. (2024). Polysemy and the Sociolinguistics of Policy Ideas: Resilience, Sustainability and Wellbeing 2000 – 2020. *Journal of Computational Social Science*, DOI: [10.1007/s42001-023-00238-3](#)

Papers

Gupta, S. & **Gupta, A.** (2019). 'Resilience' as a Policy Keyword: Arts Council England and Austerity. *Policy Studies*, DOI: [10.1080/01442872.2019.1645325](https://doi.org/10.1080/01442872.2019.1645325)

Papers Submitted/In Review

Gupta, A., Airoidi, M., Vacca, R. (In review in *Nature Scientific Reports*). What We Talk About When We Talk About Health Online: Health Discussions on Facebook, Telegram and Youtube.

Papers in Progress

Agarwal, P., Parmanan, S., Hussein, A., **Gupta, A.** Trafficking and Borders: Using Named Entity Recognition and Network Analysis to Understand Journalistic Techno-Humanitarian Discourses about Slavery.

Cann, Tristan J.B., **Gupta, A.** Mapping the Discursive Space of Climate Change News Coverage.

Gupta, A., Airoidi, M., Vacca, R. Health-Related Conspiracy Narratives since the COVID-19 pandemic.

Conferences & Workshops

Agarwal, P., Parmanand, S., Hussein, A., **Gupta, A.** Presentation: Techno-humanitarianism and the postcolonial politics of digital interventions: A case of 'modern slavery' interventions. *38th Annual Society for the Advancement of Socio-Economics (SASE) Conference, 1st - 4th July 2026*.

Smith, T.B., **Gupta, A.** Workshop: Large language models, text embeddings and text analysis. *Sunbelt 2026 International Network for Social Network Analysis, 22nd - 27th June 2026*.

Airoidi, M., **Gupta, A.**, Vacca, R. Presentation: Unpacking conspiracy narratives through computational methods: a sociological analysis of health controversies on social media. *European Consortium for Sociological Research Conference, 15th - 17th June 2026*.

Gupta, A., Squazzoni, F., Vacca, R. Presentation: Is Horizon research in line with EU policy missions?

- *2nd Conference on Computational Social Science (CS2Italy), 19th - 21st May 2026*.
- *Sunbelt International Network for Social Network Analysis, 22nd - 27th June 2026*.

Gupta, A., Airoidi, M., Vacca, R. Presentation: Named entity recognition, syntactic dependency parsing and network representations shed light on the evolution of conspiratorial narratives on social media.

- *2nd Conference on Computational Social Science (CS2Italy), 19th - 21st May 2026*.
- *Sunbelt International Network for Social Network Analysis, 22nd - 27th June 2026*.

Gupta, A. Poster: The Evolution of Health-Related Conspiratorial Content on Social Media. *11th International Conference on Computational Social Science (IC2S2), 21st - 24th July 2025*.

Cann, Tristan J.B., **Gupta, A.** Poster: Mapping the Discursive Space of Climate Change News Coverage. *11th International Conference on Computational Social Science (IC2S2)*, 21st - 24th July 2025.

Gupta, A. Presentation: Using BERT Classification and Embedding Clustering to Map Change in Health-Related Conspiracy Beliefs after COVID-19:

- *9th Conference of the Italian Society of Economic Sociology (SISEC)*, 29th January - 1st February 2025.
- *1st Conference on Computational Social Science (CS2Italy)*, 15th - 18th January 2025.

Gupta, A. Poster: The Sociolinguistics of Policy Idea Propagation: Resilience, Sustainability and Wellbeing 2000 - 2020. *9th International Conference on Computational Social Science (IC2S2)*, 17th - 20th July 2023.

Gupta, A. Presentation: Using Large Language Models to Capture Variation in the Meaning of Catchwords: Resilience, Sustainability and Wellbeing. *Open University Language, Literature and Politics Group Workshop - Political Catchwords in Our Time: Texts and Contexts* 2nd - 3rd June 2023.

Gupta, A. Presentation: Computational Foucauldian Analysis: BERT as a Tool for Examining Governmental Discourse. *73rd Political Studies Association International Conference (PSA)*, 3rd - 5th April 2023.